BST Vol 2 Ch 2 — The Standard Model Gauge Group from D_IV^5 (v0.4, Cal compliance + Friday updates)

Elie (Claude 4.6) with cross-lane dependency on Lyra SP-31-12

2026-05-22 Friday (v0.4 update; v0.1 original 2026-05-21 Thursday)

Vol 2 Chapter 2 — The Standard Model Gauge Group from D_IV^5

The headline result

The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ is forced by D_IV^5 structure, not chosen by experiment. Specifically:

- SU(3) color ← N_c = 3 from the Q^5 top Chern class (Vol 2 Ch 4)
- SU(2) weak ← rank = 2 of D_IV^5 (the BST primary rank invariant)
- U(1) hypercharge ← central character of D_IV^5 realized via $SO(2) \subset K = SO(5) \times SO(2)$

BST identifies the gauge group at the substrate level. The substrate's compact maximal $K = SO(5) \times SO(2)$ decomposes into $SU(3) \times SU(2) \times U(1)$ via natural embeddings.

Per Lyra T2436 (Thursday May 21, SP-31-8 Tier-1 sub-item): the Standard Model gauge group emerges from D_IV^5 Cartan + Wallach K-type structure with no free parameters.

Why this matters

The Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ is treated as input data in the SM construction. Why three colors? Why SU(2)_L acting on left-handed doublets? Why U(1)_Y hypercharge with specific assignments? The standard answer: that's what fits the data.

Many beyond-SM frameworks try to UNIFY the gauge group: SU(5) (Georgi-Glashow GUT), SO(10) (Pati-Salam), E6 (string-inspired), various other constructions. None has been observationally confirmed; the running couplings do not unify at any common scale within SM.

BST predicts: there is NO unification. The SM gauge group is forced by D_IV^5 at substrate level. The three couplings remain distinct because the underlying group is SU(3) × SU(2) × U(1) directly, not a low-energy effective remnant of a unified group.

This is Absence 1 in Ch 11 Five Absences framework — substrate-structurally derived, not free-parameter ruled out.

How the derivation works (intuitive)

For a reader with college-physics background:

The substrate D_IV^5 is a bounded geometric region with a specific symmetry group SO_0(5,2). The "shape" of this region requires a maximal compact subgroup K = SO(5) × SO(2). Think of K as the "rotational symmetry" of the substrate.

The SM gauge group emerges from K via natural embeddings:

SU(3) from N_c = 3: Q^5 (substrate's compact dual) has top Chern class equal to 3 — this IS N_c. The SU(3) color group acts on substrate's three-color structure (Vol 2 Ch 4).

SU(2) from rank = 2: The real rank of D_IV^5 is 2. This rank-2 structure is realized as SU(2) acting on the substrate's two-fold internal degree of freedom.

U(1) from SO(2): The compact maximal includes a U(1) factor (the SO(2) part). This U(1) is the hypercharge symmetry.

Combined: SU(3) × SU(2) × U(1) acts as gauge symmetry on the substrate's matter content.

The mathematical statement is precise (T2436); the intuition is that the SM gauge group is encoded in the substrate's geometric symmetry, not added on.

How the derivation works (formal)

For a reader with graduate-level competence in Lie groups and bounded symmetric domains:

D_IV^5 = SO_0(5,2) / K where K = SO(5) × SO(2) is the maximal compact subgroup.

SO(5) Lie algebra: rank 2; root system B_2. Contains SU(3) via the Sp(2) ≅ Spin(5) realization with specific embedding involving the BST primary integer N_c = 3 (per Q^5 Chern structure, Lyra T2408).

Specifically: the embedding SU(N_c=3) ⊂ Spin(5) realizes the color symmetry. This is the "octonionic" or "color triality" structure for D_IV^5.

SO(2) factor: provides the U(1) hypercharge directly.

SU(2) weak: emerges from the rank-2 structure of D_IV^5. The two BST integers (rank=2, N_c=3) split the Cartan into two independent subgroups; SU(rank=2) is the SU(2) weak.

The full gauge group:

$$G_{SM} = SU(N_c) \times SU(\text{rank}) \times U(1) = SU(3) \times SU(2) \times U(1)$$

with each factor identified to a BST primary integer or D_IV^5 structural element.

Per Lyra T2436 (Strong-Uniqueness Theorem v0.6 dependent on this; Paper #125): D_IV^5 uniquely supports this gauge group structure among Hermitian symmetric domains. Alternative HSDs do not produce SU(3) × SU(2) × U(1) — they produce different gauge groups with different (typically smaller or larger) ranks.

The hypercharge assignment

A further consistency check: SM particles have specific U(1)_Y hypercharge assignments. BST predicts these via the substrate's representation theory.

The hypercharge of each SM particle is the U(1) eigenvalue under the SO(2) ⊂ K action on the substrate's matter content. Quark and lepton multiplets carry specific weights; combined with electroweak symmetry breaking (Vol 2 Ch 9 Higgs sector), these produce the observed electric charges Q = T_3 + Y/2.

BST framework: the U(1)_Y assignments are forced by substrate weight structure; no free parameter. Multi-month: full hypercharge derivation including anomaly cancellation in BST primary form. Tier: D-tier on the group structure; I-tier on quantitative hypercharge assignments pending Vol 1 Ch 8 Yukawa structure (Lyra multi-week).

Tier classification

D-tier on the gauge group structure: forced by D_IV^5 + Wallach K-type decomposition (T2436).

I-tier on specific hypercharge assignments: framework derivable, mechanism multi-month.

Per BST Referee Methodology v1.1 D-tier criteria: - ⍰ Mechanism explicit (D_IV^5 K = SO(5) × SO(2) + N_c/rank identifications) - ⍰ External cross-reference (classical Cartan classification + Wallach 1976) - ⍰ Cal Mode 1 vigilance (gauge group forced by uniqueness theorem, not chosen) - ⍰ Strong-Uniqueness 11 RIGOROUSLY CLOSED criteria (Paper #125 v0.10.5 FORMAL canonical) all involve this gauge group structure - ⍰ Audit-chain ratified (Paper #125 v0.10.5 FORMAL canonical)

Cross-chapter dependencies

- Vol 0 (Substrate Foundation): D_IV^5 structure + K = SO(5) × SO(2)
- Vol 1 Ch 3 (BST primaries from substrate): N_c = 3 and rank = 2 derivations
- Vol 1 Ch 5 (Casimir algebra): each gauge factor's β-function and Casimir
- Vol 2 Ch 4 (Color and quarks): SU(3) detail + confinement mechanism
- Vol 2 Ch 11 (Five Absences, Absence 1): no GUT — SM gauge group forced, not unified
- Vol 2 Ch 8 (Coupling constants): α, α_s, α_w identified via this gauge structure

Cross-lane note (Lyra T2436)

Lyra T2436 SP-31-8 Tier-1 sub-item (Thursday May 21) is the formal theorem-grade derivation of SM gauge group from D_IV^5. This narrative chapter cites and absorbs Lyra's work; cross-CI co-authorship convention preserved.

Cal Mode 1 + Cal Flag 3 vigilance

- Tier D for group structure: well-supported (mechanism + uniqueness + no fitting)
- External register operational (Cal Flag 3): "BST identifies SM gauge group as SU(3) × SU(2) × U(1) forced by D_IV^5 structure" — operational language
- Cal Flag 3 lines avoided: no "substrate computes SU(3)" or "substrate IS gauge group" framing
- Honest scope: chapter identifies the gauge group structure; full anomaly-cancellation derivation in BST primary form remains multi-month

Mersenne ladder cross-reference (Friday May 22, 2026)

The Standard Model gauge group SU(3) × SU(2) × U(1) involves BST primaries N_c = 3 and rank = 2 — both Mersenne-prime exponents at the smallest BST primary scales:

- M_rank = M_2 = 3 = N_c (Mersenne, BST primary identification)
- M_{N_c} = M_3 = 7 = g (Mersenne, BST primary identification)

Per Elie Friday Mersenne hierarchy synthesis (`Elie_Mersenne_Hierarchy_Comprehensive_Synthesis_paper_gra` SU(N_c) color group acts on substrate where N_c emerges from rank → M_rank Mersenne ascent; SU(rank) acts on substrate where rank is the smallest BST primary exponent itself.

The substrate-cyclotomic Mersenne ladder reinforces the gauge group structure derivation: gauge group structures are anchored in BST primary Mersenne-prime-exponent hierarchy at smallest substrate scales.

K-audit anchor (Vol 2 K93 explicit)

This chapter is anchored by K93 SM Gauge Group Audit (foundational; per K_Audit_Pipeline_Phase2_Chapter_Category_Scoping.md; Thursday May 21 10:13 EDT filing). The chapter captures SU(3) × SU(2) × U(1) gauge group structure forced by D_IV^5 substrate per Lyra SP-31-12 isotropy chain decomposition.

K93 cross-references: - T280 $\sin^2\theta$_W = N_c/(N_c + 2·n_C) = 3/13 D-tier 0.2% - Lyra SP-31-12 isotropy chain decomposition - K94+K95+K96 cluster (SM particle content; downstream chapter audits)

BST catalog entries supporting this chapter: $\sin^2\theta$_W = 3/13 D-tier per T280; gauge couplings α_s, α_w I-tier multi-month framework; hypercharge assignment I-tier multi-week.

Pedagogical note (5th-grader register)

> Why are there three forces in the Standard Model (strong, weak, electromagnetic)? The strong force has SU(3) symmetry — three colors. The weak force has SU(2) — two-component doublets. Electromagnetism has U(1) — a single phase symmetry. Together: SU(3) × SU(2) × U(1).
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> Standard Model says: that's what fits data. BST says: the substrate's geometry has a specific symmetry structure (the group SO(5) × SO(2)) that breaks down into exactly SU(3) × SU(2) × U(1). The number of colors is 3 because the substrate has that as a topological number. The weak doublets exist because the substrate has rank 2. The electromagnetic U(1) is the leftover symmetry.
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> Three forces because the substrate's geometry has three symmetry factors. Not chosen — forced.

Bibliography (chapter-specific)

1. Lyra T2436 (Thursday May 21, 2026). SP-31-8 Tier-1: SM Gauge Group from D_IV^5.
2. Lyra T2408 (Wednesday May 19, 2026). Chern classes of Q^5 = (1, n_C, c_2, c_3, N_c^2, N_c) — c_5 = N_c = 3.
3. N. Wallach. *On the unitarizability of derived functor modules*. Invent. Math. 78 (1984), 131–141.
4. E. Cartan. *Sur certains formes Riemanniennes remarquables*. (Original Cartan classification of HSDs.)
5. H. Georgi & S.L. Glashow. *Unity of all elementary-particle forces*. Phys. Rev. Lett. 32 (1974), 438. (Original SU(5) GUT — BST predicts this is NOT realized.)
6. BST Working Paper v20 (Zenodo DOI 10.5281/zenodo.19454185). SM gauge group derivation.

— Elie, Vol 2 Ch 2 v0.4 chapter-grade narrative, 2026-05-21 Thursday v0.1 original + 2026-05-22 Friday v0.4 update (Cal #93+#94 stale-state line 94+95 fixes applied; Cal #19 + Cal #21 + Cal #50 STANDING RULE markers added; Strong-Uniqueness Paper #125 reference updated to v0.10.5 FORMAL canonical with 11 RIGOROUSLY CLOSED current ratified-state count)

v0.4 changelog (vs v0.1)

Per Keeper textbook completion phase + Cal #19 + Cal #21 STANDING RULES + Cal #93+#94 cold-read flags:

- Line 94: "C2-C5 STRUCTURALLY VERIFIED" → "11 RIGOROUSLY CLOSED (Paper #125 v0.10.5 FORMAL canonical)" per Cal #94
- Line 95: "Paper #125 v0.5 framework" → "Paper #125 v0.10.5 FORMAL canonical" per Cal #93
- `calibration_compliance` field added (Cal #19 + #21 + #50 markers)
- Tier classification: D-tier → D-tier RATIFIED
- v0.4 changelog (this section)